

Question 1

While researching a topic, a student has taken the following notes:

- A commodity chain is the series of links connecting the production and purchase of a commodity on the world market.
- Chinese American anthropologist Anna Tsing studies the contemporary commodity chain of matsutake mushrooms.
- At one end of the matsutake chain are mushroom pickers in Oregon.
- At the other end are wealthy consumers who buy the costly matsutake in Japan.
- According to Tsing, “Japanese traders began importing matsutake in the 1980s, when the scarcity of matsutake in Japan first became clear.”

The student wants to provide an overview of the matsutake commodity chain. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The contemporary matsutake commodity chain has its origins in the 1980s when, according to Tsing, “the scarcity of matsutake in Japan first became clear.”
- B. Commodity chains include the linked production and purchase of commodities, such as the matsutake mushroom, on the world market.
- C. Decades after the Japanese import of matsutake began, a commodity chain now links matsutake pickers in Oregon with wealthy consumers of the costly mushrooms in Japan.
- D. Wealthy consumers who buy the costly mushrooms in Japan are at one end of the matsutake commodity chain.

Question 2

While researching a topic, a student has taken the following notes:

- African American women played prominent roles in the Civil Rights Movement, including at the famous 1963 March on Washington.
- Civil rights activist Anna Hedgeman, one of the march’s organizers, was a political adviser who had worked for President Truman.
- Civil rights activist Daisy Bates was a well-known journalist and advocate for school desegregation.
- Hedgeman worked behind the scenes to make sure a woman was included in the lineup of speakers at the march.
- Bates was the sole woman to speak, delivering a brief but memorable address to the cheering crowd.

The student wants to compare the two women’s contributions to the March on Washington. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Hedgeman and Bates contributed to the march in different ways; Bates, for example, delivered a brief but memorable address.
- B. Hedgeman worked in politics and helped organize the march, while Bates was a journalist and school desegregation advocate.
- C. Although Hedgeman worked behind the scenes to make sure a woman speaker was included, Bates was the sole woman to speak at the march.
- D. Many African American women, including Bates and Hedgeman, fought for civil rights, but only one spoke at the march.

Question 3

While researching a topic, a student has taken the following notes:

- Ulaanbaatar is the capital of Mongolia.
- The city's population is 907,802.
- Ulaanbaatar contains 31.98 percent of Mongolia's population.
- Hanoi is the capital of Vietnam.
- The city's population is 7,781,631.
- Hanoi contains 8.14 percent of Vietnam's population.

The student wants to emphasize the relative sizes of the two capitals' populations. Which choice most effectively uses information from the given sentences to emphasize the relative sizes of the two capitals' populations?

- A. Mongolia's capital is Ulaanbaatar, which has 907,802 people, and Vietnam's capital is Hanoi, which has 7,781,631 people.
- B. Comparing Vietnam and Mongolia, 7,781,631 is 8.14 percent of Vietnam's population, and 907,802 is 31.98 percent of Mongolia's.
- C. Even though Hanoi (population 7,781,631) is larger than Ulaanbaatar (population 907,802), Ulaanbaatar accounts for more of its country's population.
- D. The populations of the capitals of Mongolia and Vietnam are 907,802 (Ulaanbaatar) and 7,781,631 (Hanoi), respectively.

Question 4

While researching a topic, a student has taken the following notes:

- The *Atlantic Monthly* magazine was first published in 1857.
- The magazine focused on politics, art, and literature.
- In 2019, historian Cathryn Halverson published the book *Faraway Women and the "Atlantic Monthly."*
- Its subject is female authors whose autobiographies appeared in the magazine in the early 1900s.
- One of the authors discussed is Juanita Harrison.

The student wants to introduce Cathryn Halverson's book to an audience already familiar with the *Atlantic Monthly*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Cathryn Halverson's *Faraway Women and the "Atlantic Monthly"* discusses female authors whose autobiographies appeared in the magazine in the early 1900s.
- B. A magazine called the *Atlantic Monthly*, referred to in Cathryn Halverson's book title, was first published in 1857.
- C. *Faraway Women and the "Atlantic Monthly"* features contributors to the *Atlantic Monthly*, first published in 1857 as a magazine focusing on politics, art, and literature.
- D. An author discussed by Cathryn Halverson is Juanita Harrison, whose autobiography appeared in the *Atlantic Monthly* in the early 1900s.

Question 5

While researching a topic, a student has taken the following notes:

- One of history's greatest libraries was the House of Wisdom in Baghdad, Iraq.
- It was founded in the eighth century with the goal of preserving all the world's knowledge.
- Scholars at the House of Wisdom collected ancient and contemporary texts from Greece, India, and elsewhere and translated them into Arabic.
- Writings included those of the Greek philosopher Aristotle and the Indian mathematician Aryabhata.
- The House of Wisdom used Chinese papermaking technology to create paper versions to be studied and shared.

The student wants to explain how the House of Wisdom preserved the world's knowledge. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The House of Wisdom was known for bringing together knowledge from around the world, including from Greece, India, and China.
- B. Founded in Iraq in the eighth century, the House of Wisdom employed many scholars as translators.
- C. Writings from the Greek philosopher Aristotle and the Indian mathematician Aryabhata were preserved at the House of Wisdom.
- D. The House of Wisdom collected writings from different countries and created paper versions in Arabic to be studied and shared.

Question 6

While researching a topic, a student has taken the following notes:

- The magnificent frigatebird (*fregata magnificens*) is a species of seabird that feeds mainly on fish, tuna, squid, and other small sea animals.
- It is unusual among seabirds in that it doesn't dive into the water for prey.
- One way it acquires food is by using its hook-tipped bill to snatch prey from the surface of the water.
- Another way it acquires food is by taking it from weaker birds by force.
- This behavior is known as kleptoparasitism.

The student wants to emphasize a similarity between the two ways a magnificent frigatebird acquires food. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. A magnificent frigatebird never dives into the water, instead using its hook-tipped bill to snatch prey from the surface.
- B. Neither of a magnificent frigatebird's two ways of acquiring food requires the bird to dive into the water.
- C. Of the magnificent frigatebird's two ways of acquiring food, only one is known as kleptoparasitism.
- D. In addition to snatching prey from the water with its hook-tipped bill, a magnificent frigatebird takes food from other birds by force.

Question 7

While researching a topic, a student has taken the following notes:

- The National Congress of American Indians (NCAI) was founded in 1944 by representatives of fifty tribal governments.
- The NCAI was created to protect the sovereignty of Indigenous tribes.
- Napoleon B. Johnson (Cherokee) was the NCAI's first president.
- In 1975, the US Congress passed the Indian Self-Determination and Education Assistance Act (Public Law 96-638).
- This legislation formally acknowledged tribes' right to self-governance.
- The advocacy of the NCAI was a key factor in the law's passing.

The student wants to identify an accomplishment of the NCAI. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The NCAI, founded by representatives of fifty tribal governments, had Napoleon B. Johnson (Cherokee) as its first president.
- B. Founded in 1944, the NCAI was created by representatives of tribal governments from fifty sovereign Indigenous tribes.
- C. The NCAI's advocacy was key to the passing of Public Law 96-638, legislation formally acknowledging Indigenous tribes' right to self-governance.
- D. In 1975, the NCAI passed the Indian Self-Determination and Education Assistance Act, which was created to protect the sovereignty of Indigenous tribes.

Question 8

While researching a topic, a student has taken the following notes:

- Phobetor, a name drawn from Greek mythology, is an exoplanet that orbits the star PSR B1257+12, also known as Lich.
- Phobetor's mass is 0.01 times that of Jupiter, or 0.01 Jupiter masses.
- Mastika, which means "gem" or "jewel" in Malay, is an exoplanet that orbits the star HD 179949, also known as Gumala.
- Mastika's mass is 0.92 Jupiter masses.

The student wants to make and support a generalization about exoplanets. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Exoplanets that are named Phobetor orbit Lich, and those that are named Mastika orbit Gumala.
- B. Even though Phobetor and Mastika are both exoplanets, their masses are different: Phobetor's mass is 0.01 Jupiter masses, and Mastika's is 0.92 Jupiter masses.
- C. Many stars have both a designation and a proper name; for instance, PSR B1257+12 is also known as Lich, and HD 179949 is also known as Gumala.
- D. Exoplanet names have diverse origins, a fact that can be seen in the cases of Phobetor, a name drawn from Greek mythology, and Mastika, which means "gem" or "jewel" in Malay.

Question 9

While researching a topic, a student has taken the following notes:

- In a 2023 study, environmental scientist Jazmin Locke-Rodriguez and colleagues tested the use of floating treatment wetlands (FTWs) in Florida.
- FTWs are artificial floating platforms of plants used to remediate polluted or nutrient-imbalanced water.
- Finding: FTWs using marigold flowers removed 52% more total phosphorus than the control.
- Finding: The test yielded 65 market-quality blooms per square meter.
- The authors concluded marigolds showed “promising potential as a commercially viable remediating crop cultivated on FTWs in South Florida.”

The student wants to present the findings of the study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The authors concluded that marigolds grown on FTWs were “commercially viable,” having produced 65 blooms per square meter of market-quality blooms in a 2023 study.
- B. In a 2023 study, Locke-Rodriguez and colleagues found that marigolds cultivated on FTWs produced 52% more market-quality flower blooms than the control.
- C. Locke-Rodriguez and colleagues found that FTWs using marigolds not only helped remove phosphorus from the water but also yielded market-quality blooms.
- D. FTWs using marigolds, Locke-Rodriguez and colleagues found, yielded 65 flower blooms and removed 52% of phosphorus from the water.

Question 10

While researching a topic, a student has taken the following notes:

- Two opposing theories of vision divided scholars for many centuries.
- The ancient Greek mathematician Euclid (circa 300 BCE) supported the extramission theory.
- This theory held that the eyes emit a form of radiation that illuminates objects in its range.
- The ancient Greek philosopher Aristotle (384–322 BCE) supported the intromission theory.
- This theory held that objects emit a form of radiation that reaches the eyes.
- In the eleventh century, Arab mathematician Ibn al-Haytham (965–1040 CE) largely settled the debate with the first conclusive experiments supporting intromission.

The student wants to provide a historical overview of the two theories. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Scholars were divided between the extramission and intromission theories of vision until Ibn al-Haytham's eleventh-century experiments largely settled the debate in support of intromission.
- B. Through two opposing theories of vision—extramission and intromission—Euclid, Aristotle, and Ibn al-Haytham held that a form of radiation is emitted either from objects or from the eyes.
- C. While Ibn al-Haytham largely settled the debate in the eleventh century, Aristotle supported the theory of intromission centuries before.
- D. Before the eleventh century, the ancient Greek philosopher Aristotle supported the intromission theory, which held that objects emit a form of radiation that reaches the eyes.

Question 11

While researching a topic, a student has taken the following notes:

- British scholar Robert Plot described fossilized dinosaur bones in his 1676 book *The Natural History of Oxfordshire*.
- Plot earned a reputation for being the first person to have discovered dinosaur remains.
- In 1990, archaeologists in Lesotho, in southern Africa, discovered a fossilized phalanx of a *Massospondylus carinatus* dinosaur in a cave once inhabited by humans.
- Indigenous Khoesan and Basotho peoples had inhabited the cave beginning around 1100 CE.
- According to paleontologist Julien Benoit, these peoples may have found the phalanx and brought it to the cave centuries before Plot's descriptions.

The student wants to emphasize the significance of the 1990 discovery to Plot's reputation. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Benoit challenged Plot's reputation for being the first person to have discovered *M. carinatus* remains.
- B. Evidence that Khoesan and Basotho peoples may have found an *M. carinatus* phalanx as long ago as 1100 CE suggests that Plot may not have been the first person to have discovered dinosaur remains.
- C. According to Benoit's analysis of the 1990 discovery, Indigenous peoples in southern Africa may have brought the fossilized phalanx to the cave as long ago as 1100 CE.
- D. In 1990, more than three centuries after Plot claimed in his book that he had found fossilized dinosaur bones, archaeologists uncovered evidence in southern Africa that disproved his claims.

Question 12

While researching a topic, a student has taken the following notes:

- Digital Light Synthesis (DLS) is a form of additive manufacturing that utilizes light to rapidly cure liquid resin into high-quality, 3D objects.
- Step 1: Ultraviolet (UV) light images are projected up into a pool of liquid resin, where the object's first layer takes shape.
- Step 2: The partially cured resin object is raised, leaving a thin space (a "dead zone") beneath it for oxygen and liquid resin to flow through.
- Step 3: The UV light passes through the dead zone—maintaining the flow of resin—and partially cures additional layers of the object.
- Step 4: When the resin object is complete, it is baked in an oven to complete the curing.

The student wants to describe how DLS cures 3D objects. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. DLS is a form of additive manufacturing that creates a "dead zone" in which UV light solidifies layer by layer before being baked in an oven, creating a high-quality, 3D object.
- B. DLS cures 3D objects by passing through a "dead zone," adding layers to the object, then curing the object in an oven.
- C. In DLS, UV light images are projected into a liquid resin pool to cure a 3D object layer by layer; once solidified, the object is baked in an oven.
- D. In DLS, UV light is projected into layers of liquid resin until the resin solidifies and passes through a "dead zone," wherein the curing is completed.